1. Clinical Question & Core Hypothesis

Clinical Question

In HR+/HER2– breast cancer, why do stroma-rich (high TSR) tumors have higher recurrence risk despite similar clinicopathologic features? The key question is whether the histologic stromal abundance reflects an active, pro-survival extracellular matrix program that mechanistically promotes recurrence. 2) Tier 1 paths chosen: I focused on ECM/matrisome-driven mechanisms (NABA_CORE_MATRISOME genes) because TSR is an anatomy/matrix-derived metric; COL1A1 and FN1 are canonical fibrillar ECM components most directly tied to desmoplasia and altered cell–matrix signaling. 3) Tier 2 evidence supporting the claim: hormone receptor signaling (ESR1/PGR) and proliferation (MKI67) can modulate tumor sensitivity to endocrine therapy and interact with ECM-mediated survival signals; additional matrisome members (COL3A1, LAMA2) plausibly reinforce a stiff, drug-impermeable stroma. 4) Why PFI as the endpoint: PFI (progression/recurrence interval) most closely maps to the clinical problem of recurrence/DFS described for HR+/HER2– disease and is available in TCGA; it tests the hypothesis that TSR-linked ECM programs shorten time to recurrence. 5) Deliberate exclusions: I excluded immune-centric explanations (e.g., TIL density) and metabolic or rarer mutational modifiers to keep the mechanistic claim tightly focused on ECM-driven tumor resilience; I also limited anchors to two matrisome genes to satisfy the focusing rule and maintain a testable mechanism?

Working Hypothesis

In HR+/HER2– breast cancer, high stromal content (high TSR) reflects an activated ECM program with elevated FN1 and COL1A1 expression, creating a desmoplastic, stiff microenvironment that enhances integrin-β1/FAK signaling in tumor cells. This signaling promotes survival and reduces sensitivity to endocrine therapy, accelerating recurrence. Consequently, tumors that are TSR-high and exhibit high FN1 and COL1A1 expression will have shorter progression-free intervals (PFI) than TSR-low or FN1/COL1A1-low tumors, independent of ESR1, PGR, MKI67, and other standard covariates. Secondary matrisome modifiers (COL3A1, LAMA2) may amplify this effect, while ESR1/PGR status and tumor proliferation (MKI67) may modulate the magnitude of ECM-driven resistance. The hypothesis is testable by (a) demonstrating a TSR × FN1/COL1A1 interaction predicting shorter PFI in clinical datasets, and (b) showing that ECM-enhanced survival signals and reduced endocrine responsiveness can be recapitulated in ECM-mimetic in vitro models and reversed with inhibitors of integrin/FAK signaling.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	FN1, COL1A1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

FN1: FN1 are canonical fibrillar ECM components most directly tied to desmoplasia and altered cell–matrix signaling.

COL1A1: COL1A1 and FN1 are canonical fibrillar ECM components most directly tied to desmoplasia and altered cell–matrix signaling.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	COL3A1	Supporting modifier from knowledge graph
Secondary	LAMA2	Supporting modifier from knowledge graph
Secondary	ESR1	Supporting modifier from knowledge graph
Secondary	PGR	Supporting modifier from knowledge graph
Secondary	MKI67	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: observational analyses using public datasets (e.g., TCGA, METABRIC) to test the TSR × FN1/COL1A1 interaction on PFI, adjusting for ESR1, PGR, MKI67 and other covariates;

Prediction 2: mediation/interaction analyses to assess the contribution of integrin/FAK pathway genes;

Prediction 3: in vitro/organotypic 3D models with ECM coatings enriched for FN1/COL1A1 to assess endocrine therapy response, matrix stiffness effects, and rescue with integrin-FAK inhibitors;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- COL3A1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LAMA2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.